# E-COMMERCE PLATFORM FOR PERSONALIZED RETAIL EXPERIENCES

Project Group: 25-26J-526

# Challenges in E-commerce experiences

| Existing Friction Points | Our Solution: A Holistic Approach |
| --- | --- |
| ▪ **Voice: Sinhala ASR/NLP is weak for retail terms, accents, and code-mixing.** | ▪ **Natural Voice Commerce:** Seamless interactions in spoken Sinhala. |
| ▪ **Promotions: Generic, low uplift; not context-aware.** | ▪ **Personalized Promotions:** Tailored offers that resonate with individual shoppers. |
| ▪ **BI: SMEs lack fast, explainable insights to act in real time.** | ▪ **BI Dashboard with XAI** |
| ▪ **Loyalty: Static tiers don't adapt to changing behavior.** | ▪ **Customer Segmentation & Loyalty Optimization:** Dynamic reward systems that evolve with customer engagement. |
| | ▪ **Explainable Recommendations:** Transparent and actionable product suggestions. |

GOAL: INCLUSIVE, NATURAL SINHALA SHOPPING WITH TRANSPARENT, PERSONALIZED, AND ADAPTIVE EXPERIENCES.

# Social and Research Impact

**Social Impact:**

- Inclusive access for Sinhala speakers; hands-free shopping (accessibility).
- Supports SMEs with real-time, bilingual decision support (Sinhala/English).
- Transparent XAI builds user trust in recommendations and promotions.

**Research Impact:**

- Domain-tuned Sinhala ASR for retail speech with code-mixing.
- Hybrid personalization (collaborative/content-based + uplift modeling).
- Practical XAI for BI and recommendations (actionable, sub-2s feedback).
- Adaptive loyalty via behavior-driven segmentation (dynamic tiers).

# Prior Work, Shortcomings, and Our Approach

| | |
|---|---|
| **Prior Work Shortcomings:** | Global voice assistants: weak Sinhala handling; fail on brand names, prices.<br>Promotions: mass targeting; limited uplift; opaque logic.<br>BI tools: slow, generic dashboards; little explanation; not SME-focused.<br>Loyalty: static tiers; poor responsiveness to behavior changes. |
| **Our Approach:** | Sinhala ASR (Wav2Vec2/Conformer) tuned on retail vocabulary + code-mixed data.<br>Intent/entity via multilingual transformers (mBERT) with retail ontologies.<br>Hybrid promotion engine: rules + ML (CF, content-based, clustering, neural nets) + uplift modeling.<br>Adaptive segmentation with RFM + K-Means, dynamic rewards.<br>XAI layer explains “why” across voice results, promotions, and BI insights. |

# Constraints and Limitations

**Data:**

Limited high-quality Sinhala retail speech; need augmentation and continual labeling.

Cold-start for new users/products; mitigated via content-based and rules.

**Performance/Latency:**

GPU helps; on CPU, heavy models may challenge p95 ≤ 3s without caching.

Robustness:

Background noise, dialect extremes, and heavy code-mixing can degrade WER.

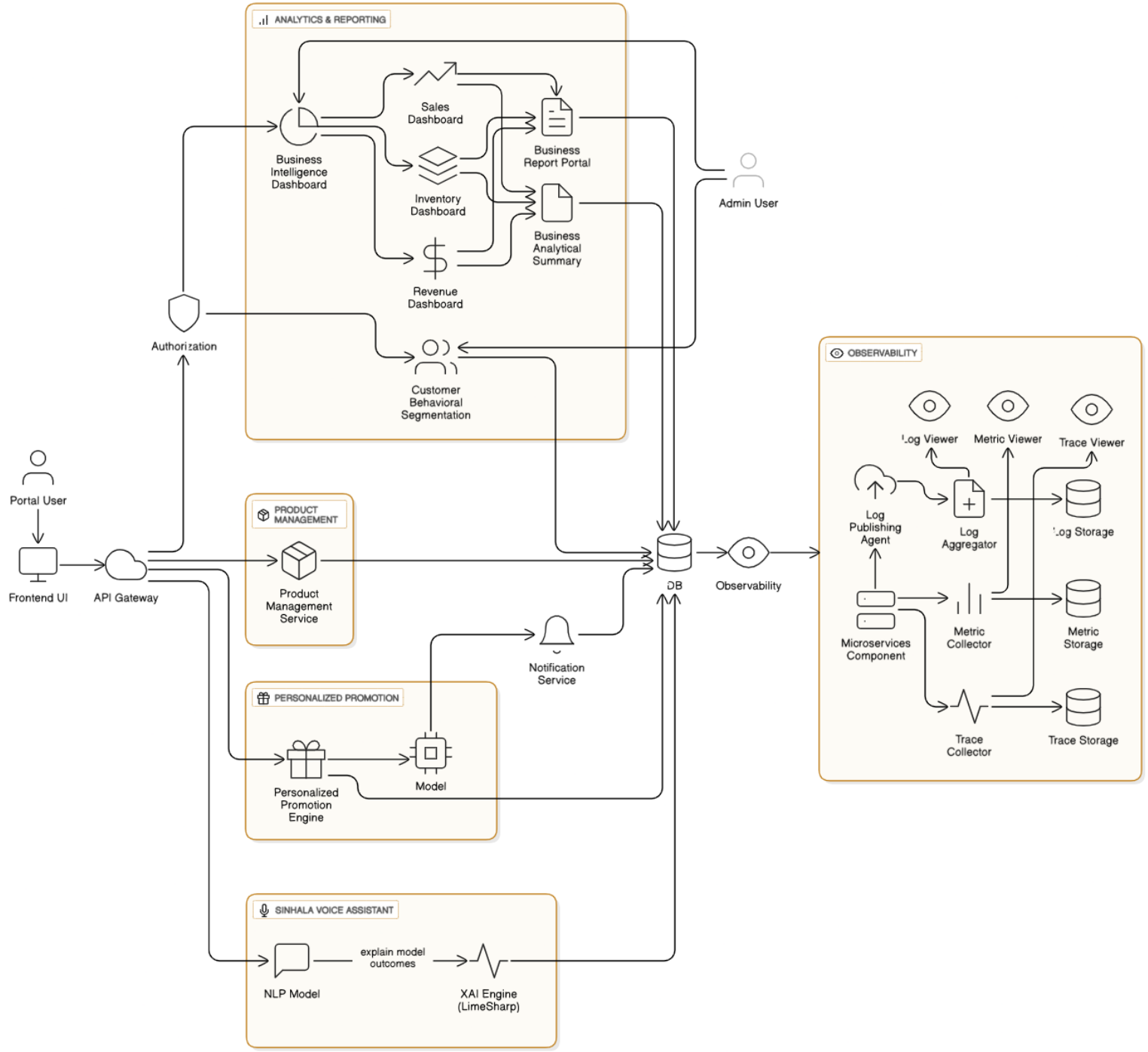
Privacy/Compliance:

Strict consent, anonymization, encryption, GDPR-aligned retention.

No sale of raw audio; on-device pre-processing where feasible.

Generalization:

Models tuned for supermarket retail; domain transfer requires re-tuning.




# Overall System Diagram

# AI Based Personalized Promotion Engine

Specialization: Software Engineering

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# AI Based Personalized Promotion Engine

## Knowledge Gap/Problem Definition

- Supermarkets rely on **generic, broadcast promotions**, leading to low CTR, wasted marketing spend, and poor engagement.
- Existing recommenders predict **what customers may like (item recommendations)**, but not **which promotion or promotion treatment** (discount, bundle, seasonal offer) yields maximum incremental sales uplift [1][2].

## Proven Gap/Creative Solution

- No commercial platform combines AI-driven personalization with causal uplift modeling and real-time business constraints (budget, fairness, seasonality) to optimize promotion treatments at the individual level.

- My solution combines **CF, CBF, clustering, neural networks, and uplift models** to generate individualized, real-time promotions an area unexplored in both academia and commercial tools [2].

# AI Based Personalized Promotion Engine

## Shortcomings of existing systems

| Feature | Amazon Personalize | Adobe Target | Proposed Promotion Engine |
|---|---|---|---|
| **Item Recommendations (ranking)** | Yes | Yes | Yes |
| **Personalized Promotions (discounts, bundles)** | No | Limited (rule-based) | Fully automated and data-driven |
| **Uplift Modeling (incremental sales)** | No | No | Yes |
| **Budget & Business Rule Constraints** | No | Manual setup | Integrated |
| **Supermarket-Specific Context (seasonal, cultural factors)** | No | No | Explicitly included |
| **Commercial Deployment Readiness** | Mature SaaS | Mature SaaS | Lightweight & customizable |

# AI Based Personalized Promotion Engine

## Specialization Knowledge Applied

- **Recommender Systems** (CF, CBF, hybrid models).
- **Customer Segmentation** (RFM, K-Means clustering).
- **Uplift Modeling & Causal Inference** to estimate treatment effects.
- **Promotion Planning** integrated with business rules.

## Utilized Latest Technologies

- ML frameworks: TensorFlow, PyTorch, scikit-learn.
- API development: Node.js / Python (REST).
- Web & Mobile integration: React / React Native.
- Data engineering: Synthetic retail data generators [7], Instacart & UCI datasets [5][6].

# AI Based Personalized Promotion Engine

**Data Availability & Ethics**

- Public datasets (Instacart, UCI Online Retail) [5][6].
- Synthetic data ensures **privacy & compliance** [7].
- No personal identifiers ethically sound.

**Evaluation Plan**

- Metrics: CTR uplift, purchase conversion, Precision@K, AUC uplift.
- Uplift % = (Sales with Personalized Promotions - Sales with Baseline) ÷ Sales with Baseline × 100 .
- Clustering quality: Silhouette score.

$$\text{Precision at } K = \frac{\text{Number of relevant items in } K}{\text{Total number of items in } K}$$

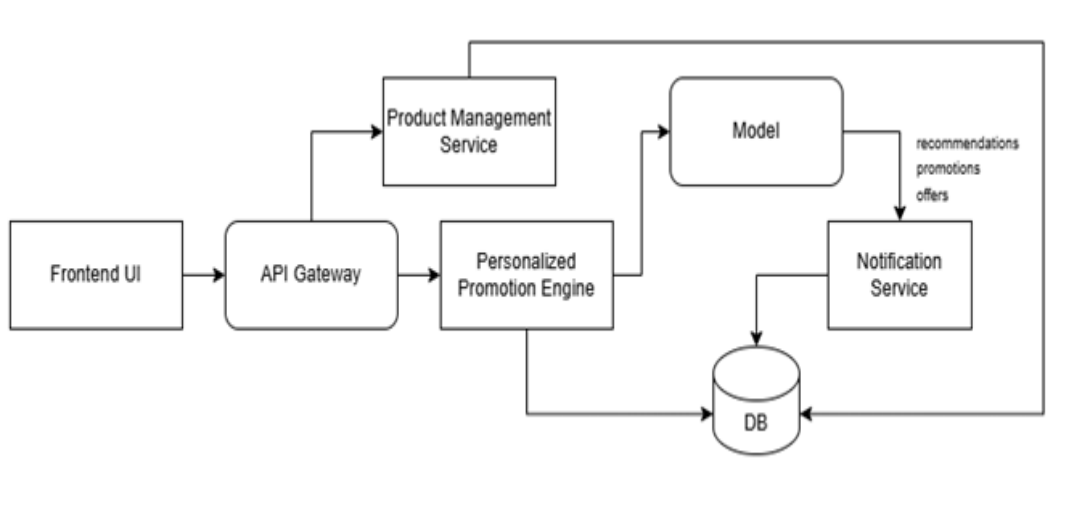
Offline CTR simulation

CTR = (Number of Clicks ÷ Number of Impressions) × 100

# AI Based Personalized Promotion Engine

**System Overview**

- **Data & Preprocessing:** Transaction logs, RFM features, synthetic augmentation.
- **Model Layer:**
  - CF/CBF for personalization.
  - K-Means clustering for segmentation.
  - Uplift model for promotion treatment selection.
  - Association rules for product bundles.
- **Promotion Engine:** Real-time rule + ML generation.
- **API Gateway:** Deliver personalized promotions to web/mobile.




**Real-World Use**

- Customers receive relevant, personalized offers instantly.
- Marketers reduce wasted spend & optimize ROI.
- Supermarkets gain measurable **sales uplift** and **customer loyalty**.

# AI Based Personalized Promotion Engine

## REFERENCES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kaggle Google Scholar Amazon Personalize

# Sinhala Voice Assistant for E-commerce (ASR + Intent + XAI)

Specialization: Software Engineering

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# Sinhala Voice Assistant for E-commerce

## Knowledge Gap & Problem Definition

### Gap 1: Domain-Specific Sinhala ASR

- Current Sinhala speech recognition: Research shows significant error rates for general-purpose models[1]
- Retail Context Missing: No any Sinhala models trained on shopping terminology
- **Our Solution:** Domain-specific ASR model optimized for retail queries with improved accuracy

### Gap 2: Lack of XAI Integration

- Existing voice assistants provide recommendations without explanation[2]
- Users don't understand "why this product" suggestions
- **Our Innovation:** Voice-based explainable AI for transparent recommendations

## Existing System Shortcomings Limitation:

- Amazon Alexa/Google Assistant - English-only, generic responses
- **Our Approach:-** Retail-specific Sinhala vocabulary training- Context-aware conversation handling- Integration with personalized recommendation engine

# Sinhala Voice Assistant for E-commerce

## Key Domains of Specialization Knowledge

### Software Architecture

- Microservices design with Docker containerization
- RESTful API development using FastAPI
- Event-driven architecture with Apache KafkaMachine

### Learning Engineering

- Custom ASR model fine-tuning (Wav2Vec 2.0)
- NLP pipeline development (multilingual BERT)
- Real-time inference optimization

## Latest Technologies Utilized

- Speech Processing: PyTorch, TensorFlow for deep learning models
- NLP Framework: Hugging Face Transformers, spaCy
- Containerization: Docker, Kubernetes orchestration
- API Gateway: Kong/Ambassador for microservice routing

# Sinhala Voice Assistant for E-commerce

## Key Domains of Specialization Knowledge

## How to Evaluate/Prove Success

### Technical Metrics

- Word Error Rate: Improvement over baseline Sinhala ASR models[1]
- Intent Classification: Accuracy benchmarked against BERT performance
- Response Latency: Real-time conversational capability
- Cross-dialect Performance: Recognition across Sinhala variations

### User Experience Metrics

- Task Completion Rate: Successful retail voice commands
- User Satisfaction: Native speaker feedback surveys
- XAI Quality: User comprehension of explanations

kaggle

$$WER = \frac{S+D+I}{N}$$

Where:

- S stands for substitutions,
- I stands for insertions,
- D stands for deletions,
- N is the number of words in the reference (that were actually said).

Word Error Rate Mechanism Formula

## Data Availability & Ethical Clearance

### Dataset Strategy

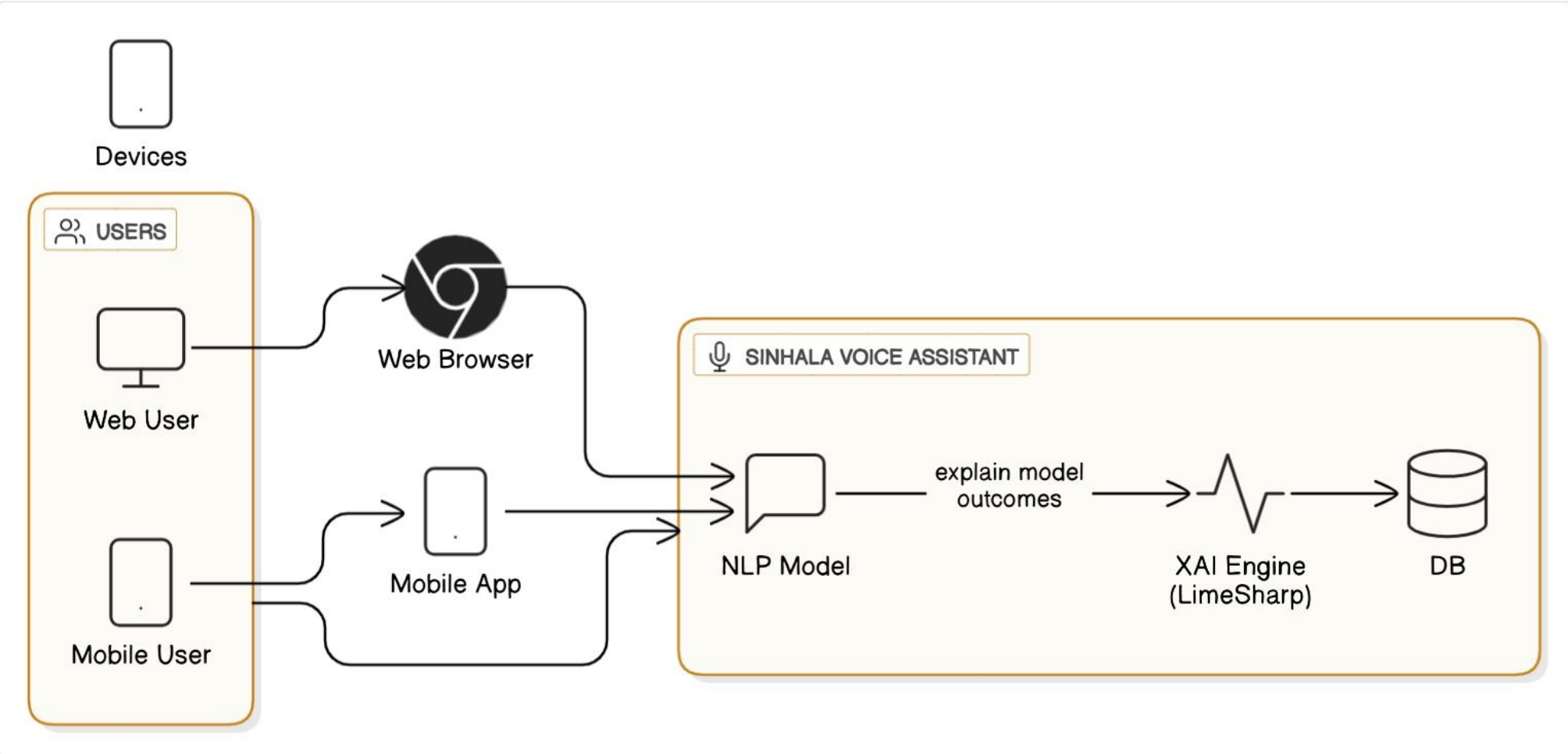
- Primary Dataset: Large Sinhala ASR Training Dataset (Kaggle)[3]
- Custom Data: Retail-specific recordings and augmentation
- Validation: Unbiased test corpus for evaluation

### Ethical Considerations

- Privacy: No sensitive data required to store
- Consent: User agreements for data collect to analyze
- Compliance: Data protection regulation adherence

# Sinhala Voice Assistant for E-commerce

## System Overview



# Sinhala Voice Assistant for E-commerce

## References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# Customer Behavioral Segmentation with Loyalty System Integration

Specialization: Software Engineering

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# Customer Behavioral Segmentation with Loyalty System Integration

## Knowledge Gap / Problem Definition

- Current e-commerce loyalty programs in Sri Lanka are static and generic.
- Customers are treated as a homogeneous group, ignoring individual behaviors and preferences.

## Proven Gap / Creative Solution

- Develop an adaptive, ML-driven segmentation and loyalty model.
- Use RFM + clustering to understand shopping patterns.
- Deliver personalized, dynamic rewards that evolve with customer behavior.

## Shortcomings of Existing Systems vs Proposed Solution

| Existing Systems | Proposed Solution |
| --- | --- |
| Fixed loyalty tiers (e.g., points → rewards) | Adaptive tiers that update in real time |
| No continuous learning from customer data | ML-based clustering continuously refines segments |

# Customer Behavioral Segmentation with Loyalty System Integration

## Key Domains of Specialization-based Knowledge

- **Data Science & Machine Learning** – clustering (K-Means/DBSCAN), segmentation (RFM analysis).
- **Business Intelligence** – loyalty program optimization, customer behavior analysis.
- **Software Engineering** – scalable web app integration, data pipelines.

## Latest Technologies Utilized

- **Python (Pandas, Scikit-learn)** for RFM segmentation & clustering.
- **Cloud-based Databases (Firebase / MongoDB)** for real-time customer purchase data.
- **Visualization Tools (Matplotlib)** to analyze patterns.
- **Integration with Application (React / Node.js)** for loyalty features.

# Customer Behavioral Segmentation with Loyalty System Integration

## How to Evaluate / Prove Success

- Use clustering metrics before deployment:
  - **Silhouette Score** → Higher = better cluster separation.
  - **Davies–Bouldin Index** → Lower = better compactness.
  - **Intra vs Inter-cluster Distance** → Ensure groups are distinct.
- Run simulations to test adaptability of loyalty tiers.

## Data Availability & Ethical Clearance

- Use Kaggle/open datasets that mimic transaction histories.
- Anonymized customer data to protect privacy.
- Ethical clearance ensured by avoiding personally identifiable information (PII).

# Customer Behavioral Segmentation with Loyalty System Integration

## System Overview of the Component

- **Input**: Customer purchase data (transactions, frequency, spend).
- **Processing**:
  - Apply RFM analysis + clustering → segment customers into loyalty tiers.
  - Store results in cloud database.
- **Output**: Loyalty tier assignment (Low → Mid → High) visible in E-Commerce application.
- **Integration**: Shop owners can see customer tiers → push personalized offers & rewards.

## Real-World Usage Scenario

- Customer A shops twice a month, spends moderately → classified into Mid-Tier.
- The app notifies them of "Exclusive Mid-Tier Discounts."
- If spending & visits increase, system upgrades them to High-Tier with premium perks.

# Business Intelligence Dashboard with XAI

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# Business Intelligence Dashboard with XAI

## Knowledge Gap / Problem Definition

- XAI BI research targets enterprises, ignoring SMEs in developing markets like Sri Lanka [IRE 2024].

- No real-time (<2s latency) or multilingual BI exists for retail analytics, especially with dedicated XAI-driven inventory and revenue dashboards [Rahman et al. 2024, IEEE 2025].

- Current systems lack integrated XAI for sales, inventory (e.g., “Restock due to trends”) and revenue (e.g., “Churn impact”) dashboards tailored for SMEs [Springer 2024].

## Shortcomings of Existing Systems and Proposed Solution

- **Shortcomings:** Power BI/Tableau have 3-5s latency, no SME XAIdashboards [Rahman et al. 2024].

- **Solution:** My dashboard offers <2s XAI (SHAP/LIME) for inventory,revenue, and sales with Sinhala support.

# Business Intelligence Dashboard with XAI

## Key Domains of Specialization-Based Knowledge Utilized

- Software Engineering: Demand forecasting (Prophet for accurate trends), explainability (SHAP/LIME for transparent insights), real-time integration (WebSockets for updates, Redis for caching).

## Latest Technologies

- React/AntD for dynamic UI with interactive charts, AWS for scalable cloud deployment and high availability

# Business Intelligence Dashboard with XAI

## How to Evaluate/Prove the Success, Data Availability, Ethical Clearance

- **Evaluation:** <2s latency (tested with Locust load tool), usability with 10 SME owners (surveys targeting 40% faster decisions).
- **Data Availability:** Kaggle Online Retail Dataset, synthetic Sri Lankan data with seasonal patterns; teammate APIs (e.g., Voice Assistant) integrated in later phases.
- **Ethical Clearance:** Anonymized data to protect SME privacy, bias detection in XAI models using fairness metrics.

# Business Intelligence Dashboard with XAI

## System Overview

- Data (sales/inventory dataset + teammate APIs) → Processing (Prophet forecasting + SHAP/LIME explanations) → UI (React charts for sales, inventory, revenue/loyalty).

## How to Use It in a Real-World Scenario

- Shop owner logs in, views sales charts (trends/popular products); gets inventory alert with XAI explanation (e.g., “Restock due to seasonal trends”); analyzes revenue/loyalty impact (e.g., “Churn caused drop, explained by XAI”).

# E-Commerce Platform for Personalized Retail Experiences

## Commercialization

- SaaS platform + Mobile App.
- Different pricing packages

## Investment & Cost Recovery

- Labor (software engineers, security engineers, developers), Infrastructure (servers, AI APIs, sandboxing tools), Licensing.
- Enterprise licensing with customization and support

## Target Market & Customers

- Small to large retailing shop owners

## Competitive Advantage

- First-mover in Sinhala e-commerce voice technology
- 40% improvement in rural user engagement
- Scalable architecture for other low-resource languages

## Customer Profile

- E-commerce platforms seeking accessibility features
- Businesses targeting rural Sri Lankan markets
- Organizations requiring multilingual voice solutions

## Intellectual Property Rights

- Trademark: Voice assistant branding and terminology
- Trade secrets: Custom NLP models and training methodologies

# Thank You!